

Logarithms: Properties, Applications & Derivatives

MATH 145 – Applied Calculus

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Logarithmic Functions

We know that if:

$$x = 2^3,$$

that $x = 8$.

But how would we solve the following equation for x ?

$$2^x = 8$$

Logarithmic Functions

Logarithm to the base b

Given an exponential function b^x , the **logarithm to the base b** is the function denoted by $\log_b(x)$ and defined by the equations

$$\log_b(b^x) = x \qquad \text{and} \qquad b^{\log_b(x)} = x$$

The logarithm is the inverse operation of exponentiation, just like how division is the inverse operation of multiplication or how subtraction is the inverse operation of addition.

Common Logarithm

Common Logarithm, $\log_{10}(x)$

$\log_{10}(x)$ is called the **common logarithm** and is generally denoted by $\log(x)$.

Base 10 is the numeral system we use in general life.

Natural Logarithm

Natural Logarithm

$\log_e(x)$ is called the natural logarithm and is typically denoted by $\ln(x)$.

Note that:

$$\ln(e^x) = x \qquad \text{and} \qquad e^{\ln(x)} = x$$

The natural logarithm is the most important logarithm and the one you will use most often.

Caution! Because it's so important and widely-used, occasionally you may see online sources use “log” to refer to “ $\log_e$ ” and not “ $\log_{10}$ ”. Make sure you read what they are doing closely. I will try my best to always use the notation described in this lecture.

Example: Logarithms

1. Solve $e^{7x} = 12$.
2. Solve $4 + 2e^{-x} = 8$.
3. Solve $10000^x + 75 = 98$.

Properties of Logarithms

Note that $\log(x)$ and $\ln(x)$ are not defined when x is negative or 0.

$$1. \log(ab) = \log(a) + \log(b).$$

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$$2. \log\left(\frac{a}{b}\right) = \log(a) - \log(b).$$

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$$3. \log(a^p) = p \log(a).$$

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$$4. \log(10^x) = x.$$

$$4. \ln(e^x) = x.$$

$$5. 10^{\log(x)} = x$$

$$5. e^{\ln(x)} = x$$

$$6. \log_b(x) = \frac{\ln(x)}{\ln(b)}. \text{ (This works for any base, not just } \ln.)$$

Example: Logarithms

1. Solve $4(2^t) - 2 = 3$.
2. Solve $3 + 2 \log(t) = 3.5$.
3. Solve $5 = 4(2^{x-1})$.

Example: Logarithms

Example: One hundred kilograms of radioactive substance decays to 40kg in 10 years.

1. How much material is left after 20 years?
2. What is the half-life of the material? How long until 50kg of the material remains?

Derivative of the Natural Logarithm

Recall: $\frac{d}{dx}e^x = e^x$, and by the Chain Rule, $\frac{d}{dx}e^{f(x)} = e^{f(x)}f'(x)$.

What is $\frac{d}{dx}\ln(x)$?

Consider $x = e^{\ln(x)}$.

$$\begin{aligned}x &= e^{\ln(x)} \\ \Rightarrow \frac{d}{dx}x &= \frac{d}{dx}e^{\ln(x)} \\ \Rightarrow 1 &= e^{\ln(x)} \cdot \left(\frac{d}{dx}\ln(x)\right) \\ 1 &= x \left(\frac{d}{dx}\ln(x)\right) \\ \Rightarrow \frac{d}{dx}\ln(x) &= \frac{1}{x}\end{aligned}$$

Derivative of the Natural Logarithm

$$\frac{d}{dx} \ln(x) = \frac{1}{x}$$

In general:

$$\frac{d}{dx} \ln(f(x)) = \frac{f'(x)}{f(x)}$$

Example: Derivative of the Natural Logarithm

1. Solve $\frac{d}{dt} \ln(3t + 1)$.

2. Solve $\frac{d}{dx} x^2 \ln(x^2 + 2)$.

3. Solve $\frac{d}{dx} \ln(2^{3x+4})$.

Derivative of the Natural Logarithm: Consequences

$$\frac{d}{dx}a^x = a^x \ln(a)$$

$$\frac{d}{dx} \log_a(x) = \frac{1}{x \ln(a)}$$

Example: Derivative of the Natural Logarithm

1. Solve $\frac{d}{dt} 2^{3x}$.

2. Solve $\frac{d}{dx} x^2 \log(3x)$.